Data Science Tools: Languages, Libraries & APIs
(MODULE 1)

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Data Science Languages Overview

In Data Science Programing language are for:

- Data manipulation & analysis
- Statistical modeling & ML
- Data visualization
- Scientific computing

Why the right choice matter:

- Faster result
- Better communication and pairing if popular
- Wider job opportunity
- Some task are easier

Python - “The Generalist”

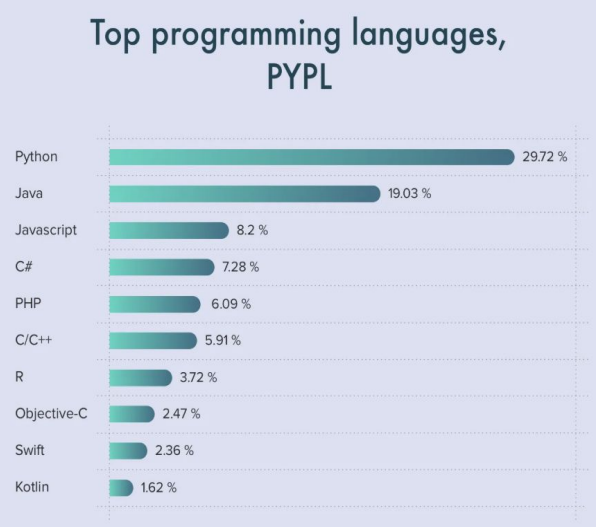
Python(1991) created for simplicity and readability. Gained popularity in the field of data science because of its powerful libraries and industry support.

<u>Best for:</u>

- Machine learning & AI
- Web development + Data
- Production system

<u>Many Libraries:</u> Pandas (data manipulation), NumPy (numerical), TensorFlow (machine learning),PyTorch (deep learning)

Top 7 Companies That Use Python Are Making A Mark In 2026



The PYPL index are base on language tutorials are searched on Google.

R - “build for datas”

R(1993) specifically created for statistical computing. Gained popularity due to its powerful data analysis and visualization tools.

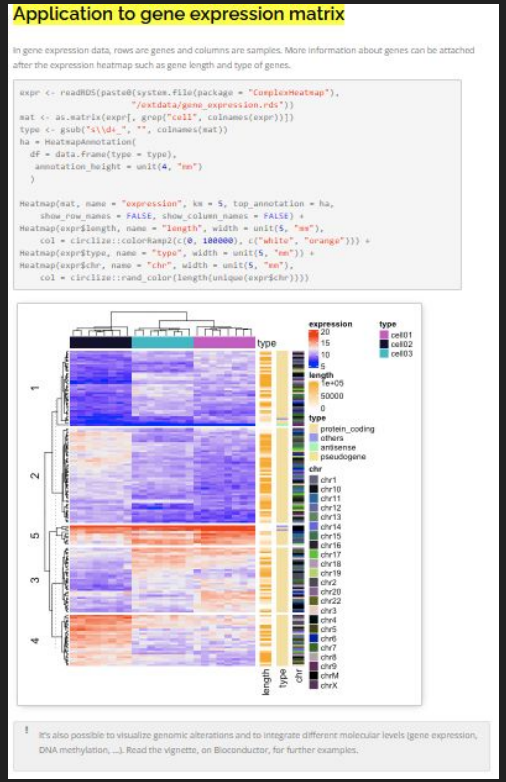
<u>Best for:</u>

- Statistical Analysis
- Data Visualization
- Academic Research

<u>Key libs:</u> ggplot2 (visualization), dplyr (data manipulation)

Some applications of R

1. UK Office for National Statistics - Produces official government statistics with R
2. Cancer Research (University of Salzburg) - Analyzes tumor genetics using R
3. Spotify - R is used by developers to analyze user listening habits
4. Netflix - Stock price prediction using machine learning in R
5. Google Play Store - Review analysis and sentiment tracking with R



R is used in cancer research to visualize gene expression patterns across different cell samples

Matlab - "Numerical Computing Perfected"

MATLAB(1984) created for numerical computing. It is widely used in the fields of engineering, physics, and mathematical simulations.

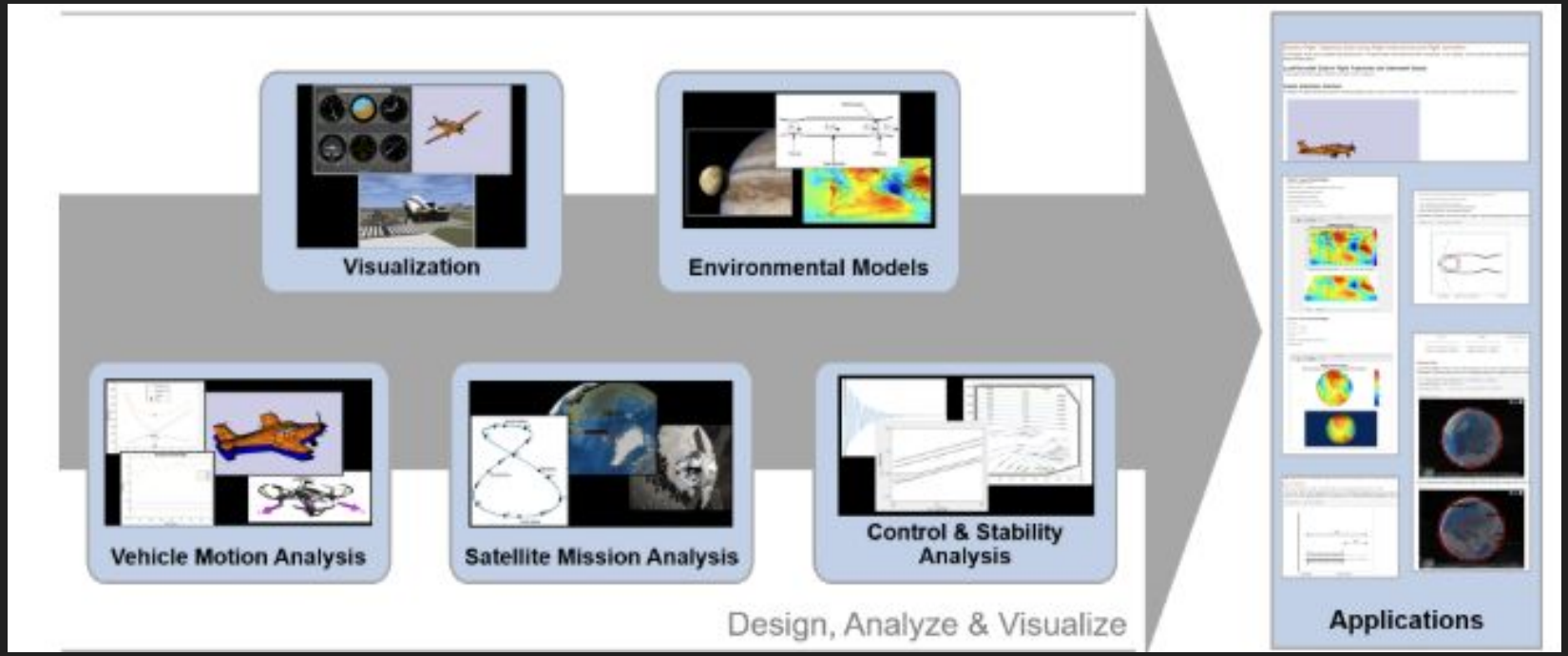
<u>Best For:</u>

- Numerical Computing
- Engineering Simulations
- Signal & Image Processing

<u>Key toolbox:</u> simulink, signal processing, control system

Wide Known Companies that uses Matlab

1. NASA: Utilizes MATLAB for data analysis, simulation, and modeling in various aerospace projects.
2. Tesla: Employs MATLAB for developing algorithms and simulations related to electric vehicle technology and autonomous driving systems.
3. Intel: For modeling circuit behavior and performance characterization.
4. Apple: In various engineering and design applications.



“**Aerospace Toolbox** provides standards-based tools and functions for analyzing the motion, mission, and environment of aerospace vehicles.” - Matlab Help Center

Web Scraping Libraries

They are tools that help you automatically extract data from websites. They read the HTML code of webpages and pull out the information you want.

Libraries	Beautifulsoup	Scrapy	selenium
Best for	Html, parsing, simple static sites	scale large projects, fast built in explore feature	Java, sites, login, clicking
Difficulty	beginner	intermediate	advanced

Example: Here, we first send an HTTP request to the webpage, then use BeautifulSoup to parse the HTML content and format it in a clean, readable structure.

```python
import requests
from bs4 import BeautifulSoup

res = requests.get('https://www.geeksforgeeks.org/python/python-programming-language-tutorial/')
soup = BeautifulSoup(res.content, 'html.parser')
print(soup.prettify())
```

Output

```
   <button class="suggest-bottom-btn" type="button">
   </button>
  </div>
 </div>
 <a class="create-improvement-redirection-to-write" href="#" style="visibility:hidden" target="_blank">
 </a>
</div>
<script>
 //code to add text editor on click in suggestion modal in improvements
          let suggestEditorInstance=null; //global variable to access editor instance
          let editorLoaded = false;
          $('.suggest-change_wrapper,.improve_dot_dropdown_inner_div, .locked-status--impove-modal .improve-bottom-btn ,#user-personal-note ').one('click',function() {
          //editor already loaded then return
          if(editorLoaded){
          return;
          }
          // if($('.spinner-loading-overlay:eq(0)').length){
          // $('.spinner-loading-overlay:eq(0)').remove();
          // }
          // $('body').append('<div class="spinner-loading-overlay"></div>');
          // $('.spinner-loading-overlay').show();
          let script = document.createElement('script');
          script.src = 'https://assets.geeksforgeeks.org/v2/editor-prod/static/js/bundle.min.js';
          script.defer = true
          document.head.appendChild(script);
          script.onload = function() {
          suggestionModalEditor() //to add editor in suggestion modal
          if(loginData && loginData.premiumConsent){
            personalNoteEditor() //to load editor in personal note
          }
          }
          script.onerror = function() {
          if($('.editorError').length){
          $('.editorError').remove();
          }
          var messageDiv = $('<div class="editorError"></div>').text('Editor not loaded due to some issues');
          $('#suggestion-section-textarea').append(messageDiv);
          $('.suggest-bottom-btn').hide();
          $('.suggestion-section').hide();
          editorLoaded = false;
          }
```

APIs - RESTful vs GraphQL

APIs, Application Programming Interfaces, the invisible backbone of modern software development. They enable applications and systems to communicate and share data efficiently.

Company application:
REST: Twitter API, Google Maps, Stripe
GraphQL: GitHub, Shopify, Facebook (they invented it!)

APIs	RESTful	GraphQL

Ex / diff	`GET /pizzas`
	`GET /pizzas/123`
	`POST /orders`
	`{`
	`"pizzaId": 123,`
	`"Address":`
	`"123 GFG Office"`
	`}`
	
	(predefined pizza and toppings)
		`query {`
		`pizza(id: 123)`
		`{`
		`size`
		`crust`
		`toppings {`
		`name`
		`}`
		`}`
		`}`

Pro & Cons	Pro: Simple, caching Con: Over/under-fetching	PRO: Ask for exactly what you need CON: Complex query language

Complete Comparison Table

Feature	Python	R	MATLAB
Ease of Learning	Easy	Moderate	Moderate
Scalability	Highly scalable for large systems and big data	Moderate	Moderate
Library Support	Many Libraries(NumPy, pandas, TensorFlow)	Strong statistical package library	Moderate(Toolbox-based)
Integration	Excellent with web apps, cloud, APIs	Limited integration(Works with fewer tools)	Mostly standalone(Works mainly by itself)
Use Case Strength	High (Machine learning, automation, data science)	High (Statistical modeling, research analysis)	High (Numerical computing, simulations)
Memory Efficiency	Moderate	Moderate	High
Flexibility (Cost)	Free	Free	Paid

Pros & Cons

Python

Pros

- Highly versatile
- Human readable
- Free(Open source)
- Strong community
- Integration

Cons

- Slower than compiled languages*
- Managing all dependencies
- Slower computation*

MatLab

Pros

- Specializes in linear algebra engineering computations
- Readable syntax
- Great visualizations

Cons

- Requires extensive prior knowledge
- EXPENSIVE

R

Pros

- Specializes in statistical data analysis*
- Free(Open source)

Cons

- Eccentric syntax
- Smaller ecosystem despite being open source
- Slower computation

Reference

- 2026 programming language ranking
- Python popularity
- Heatmap in R
- Companies that uses python
- matlab applications
- Matlab help center
- Web scrap tutor
- MatLab vs Python vs R
- API
- RESTful vs GraphQL

Q & A